Meeting with Woodcock, Sarah L-20261011_150934-Meeting Recording

11 October 2026, 02:09pm

22m 23s

Woodcock, Sarah L started transcription

Woodcock, Sarah L 0:10

Hello, I'm Sarah Woodcock and this is my machine learning individual project recorded in October 2026.

Our subject is comparative intelligence and ethics in machine learning systems. I chose the track, project track B, recommender systems, and I chose the data set Movie Lens 1M, which is available at this URL.

So there were pros and cons to this choice. In favour of it, there are a very large amount of data, lots of bits of information about the films and the users and so on. Also, the similar data set, Movie Lens 100K, was used in our Unit 8 example, so I already had some information that I could compare my results with to cheque that I was on the right track.

In comparison with the other data sets that I looked at, so I spent a long time looking at book crossing, which is reviews of books by users. And I also looked at some theatrical review data sets, but they just didn't have anything like the breadth of information about either the items or the users. The cons,

obviously of this data set are that MovieLens 100K was used in our unit 8 example, so it would be difficult to write code that hadn't already been used. And also this data set is well known and it's widely used, so it would be difficult to draw new conclusions. But I'm new to this, so I'm quite happy if I can draw the same conclusions as people who know what they're doing.

I did some exploratory data analysis. So we start out with the ratings distribution. This seems pretty fair to me. If you've watched a film and you liked it, you're probably going to rate it a three or a four. You're only going to give it A5 if you thought it was exceptional. And if you didn't make it to the end or somebody forced you to sit through it, then you're going to give it maybe a one or a two. So those those ratings look fairly okay. In terms of ratings per user, again, it's what we'd expect to see. A couple of users who've watched a lot of films and a few users who haven't watched very many and then quite a reasonable number of people in the

middle.
The number of ratings per film, again, lots of ratings for some films, very few ratings for others, and all the others sit somewhere in the middle, just like the users. And then distribution of movie genres, this is quite interesting. So drama and comedy and action are all at the top.
there. And right down at the bottom, you've got fantasy, Western, film noir. So one of my questions that came up from this is with film noir, for example, if you watch a film noir film, you're probably a film noir buff. So you're probably going to give it quite a high rating. So would it be that all of these kind of genres at the lower end of the scale would get higher ratings because it was only their fans that were watching them?
That's something that we'll look at, hopefully. And then also we looked at user demographics. So there are twice as many men watching films as women. I could probably draw some conclusions here about who's doing the housework, but I shall move on to that and keep it scientific and just assume that
men are more predisposed to watching films than women. And in terms of the age group, it's those mid-20s who seem to have all the free time available for watching the films. So this does mean that, you know, not every type of person is equally represented, but that's
That's the real world. That's how things are. So, you know, we could try and maybe cut out some of the tail ends and just concentrate on the internal information. But I'm not sure that that would give us a recommender system that would really work in the real world. So we need to work with this information, but be aware of it.
I started with a traditional machine learning model, so model-based collaborative filtering using matrix factorization, trained with stochastic, I can't even say that, stochastic gradient descent, or SGD, which was based on our Unit 8 exercise.
These are the steps that I followed. So I pre-processed the data, built some timelines, converted the user and film IDs. This is because each user and film was anonymized, has its own individual number, but those numbers weren't consecutive. So in order to keep the system on track to make it easier,
we indexed those. So if, for example, I think there are 3,706 films, so they're now indexed as naught to 3705, so that it made all the computations, etc. easier to do.
We initialised the model, trained it,
using SGD and then evaluated the predictions.
So data pre-processing, fairly straightforward, imported the data, validated it, so

checking that everything made sense, all the ratings were between 1:00 and 5:00, etc, etc. Carried out the exploratory data analysis that I've already shown you. I checked the numbers of ratings, users and films, and I calculated sparsity, which was 95.5%. Now, as I said earlier on, that means that most users
didn't rate most films. Could be an issue, but it is a real world situation. So we're going with what we've got and we'll see what happens. Then I sorted the ratings by timestamp and split the data 80-10-10 into training, validation and test.
Baseline, so we calculated some baselines, started with the global mean against the validation and the test set, and also looked at the bias baseline against both validation and the test set. Now the RMSE and the MAE, so RMSE stands for root mean squared error,
And the MAE is for the mean absolute error. And both of those, the closer they are to 0, the better your result. So the global mean here is predicting the same rating for every user in films. That's the mean average. And that's what we're using as our benchmark to see how good our model is.
Then we introduced the bias baseline, which is accounting for the users who always give low or high ratings and accounting for the films that usually receive low or high ratings. And the results are better than the global mean overall, which is what we'd expect because we've introduced that element of bias.
on both sets of data, so or on both.
types of results, the global mean and the bias baseline, are slightly better against the test data than against the validation data. So I would consider just checking for overfitting by looking further into the RMSE variations. And I mean, those results could be explained by different
time periods covered by the data. It's probably not overfitting, but it's worth checking.
The matrix factorization using stochastic gradient descent, SGD, so we used that and this gave us a lower RMSE of 0.9187, so that's the best we've had so far.
This shows us that allowing the models to discover those hidden features can improve the prediction accuracy. Also, the NDCG of this model was 0.8785 over 976 users. We'll come back to that later on.
So I built a little UI so I could have a look at my model in action. I'm going to show you some screenshots here. So there were three tabs to this, personalised recommendations, cold start demonstration and popularity bias. So I selected a user from a drop down and I chose user 647, my old friend.

And these were the films that were suggested for him. Please don't note that that predicted rating is 5.01 because it should stop at 5. So I obviously need to just recheck my workings and make sure that I make sure it stops at 5 going forwards. On the cold start demonstration,
Obviously, this is where we have no knowledge of our user at this point. So we're just saying, hey, here's some good films. And these were the five films that it came up with in that case. And then this final screen, it's a little bit small, but it's the popularity bias. So this is where we're looking at that same user, 647. We look at his personalised recommendations, which remain the same as they were before.
but also the most popular unseen films that we would recommend to him. I'm saying him because most of our users are a him. I haven't actually checked whether he's a him or not, but I'm guessing he is. So this is really interesting to see what the top unseen films are, the top films that this guy hasn't seen yet.
and it'd be interesting to know whether he'll enjoy them or not.
I then moved on to my deep learning model, neural collaborative filtering architecture, NCF. And here's what I did. So I built and trained my model. So I reused the Movie Lens data set, so I didn't need to re-import anything. It was all there in the notebook already. Created the embedding layers with 32 dimensions for each user and film. That's because there were 32 latent factors in the tradition.
model, so trying to keep the models as similar as possible to start with. Combine the embeddings, pass them through some dense layers. I'll talk about that more on my next slide. Use sigmoid output. Compiled the model using the Adam optimizer, so that's the adaptive moment estimation, which uses a moving average of previous gradients and adaptive learning rates to train. And then I trained the model with a maximum of 20 epochs using early stopping to reduce the risk of overfitting. So here's a lovely little diagram of how we did that, that architecture, the 32 sections embedding and moving on through to the rating prediction.
The model evaluation, so we can see that for the NCF model on the test data set, the RMSE was 0.9279 and the MAE was 0.7285. This is not quite as good as you'll see on the next slide as our traditional model. Also, as we work through the epochs, we see that the training RMSE goes down and down and down.
and the validation RMSE stays roughly in the same place. So the gap between the two is widening. Again, could suggest overfitting, could be something else worth looking into that to see exactly what's going on there. But it could be that the system is learning things from the training data that aren't always true in the validation data.

Looking at the RMSE and the MAE by model, so comparing all models next to each other. In grey here, we've got the global mean and the bias baseline, because those are just kind of benchmarks that we set for ourselves. And then we look at the traditional MF and the deep learning NCF.
against each other and against those baselines to see how they did. You can see it's tiny, but the traditional MF is just ahead by a tiny nose, I would say, in both instances. Also with the NDCG, the MF is 0.8785, NCF 0.8769,
So again, just a tiny win for the MF there, but only just...
That basically tells us that a more sophisticated and complex model doesn't always give better results.
Just to be interesting, I had a look at what the most rated film was, and it was American Beauty, which had 3,428 ratings. The user with the most ratings, I guessed he'd be male, he was, but I expected him to be 25 and he wasn't, he was 50. But he did watch American Beauty and he liked it a lot. The least rated film was Juno and the Paycock.
That makes me sad because I worked on a theatrical production of Juno and the Paycock and I loved it. But that only had one rating, which was from a 56 year old chap who obviously didn't want to watch it again, I guess.
I also did a quick UI so I could have a look at my NCF rating predictor in that, so test out my results. So again, I selected a user ID and I actually selected, if I just pop back to my previous slide, this user 4169, who was the one who gave us the most ratings, I had to look at him to see what he should think of American Beauty and indeed we did predict that he would give us a 4.61 rating and he gave us a 5.5 so that's pretty good.
So critically evaluating the MF versus the NCF. I looked at a few different factors, as you can see on here. So basically, MF is more simple in its architecture. NCF is more complex, which leads to more complex calculations. So
gives it the ability to take more things into account. So it's not necessarily a bad thing. It just means that it takes more thinking about. In terms of prediction accuracy in these particular models, MF1 hands down, I mean, only by a tiny bit, but in all of the factors that I looked at, MF beat NCF every time. In terms of learning capability, MF can kind of take some non-linear modelling into account, but it's NCF that really excels at that.
So where you have that kind of relationship that you need to explore, NCF is always going to win. In terms of explainability,

The fact that we can use our SHAP means that user and film biases are partly interpretable, but the latent factors, not so much. It's not quite so easy to see what's going on there. And in NCF, embeddings and hidden layers are harder to interpret. You can't really see where the decisions are coming from.
In terms of training, MF is, again, simpler and NCF means you have to make a lot more decisions. So you can do a lot more training, you can make it more flexible, but you have to make a lot more decisions. You have to spend a lot of time building that into your model.
Deployment, MF is easier and NCF takes more effort, I would say, from computer, from the humans implementing it. There's just greater complexity overall. In terms of cold start, which is where you have very little information about the film or the user to begin with, both systems
need help there. Neither of them can handle that particularly well.
Explainability. So as I mentioned earlier on, I looked at SHAP and these are my direct results out of my notebook. And we can see how this model explains to us how the rating was arrived at and the prediction of the rating was arrived at for movie 468 and user 24.
So what happens here is that a surrogate model is created, so the data is applied to a small data set and we then try to reproduce the MF output. And so when SHAP is applied to this surrogate model, that gives us
this information on how things are being calculated. In contrast, when we apply SHAP to the NCF, it can't give us quite such a breakdown. What it does show us is how much of the rating prediction has come from the film itself and how much of it has come from the user.
Ethical and bias considerations is something that we need to look at. So when we looked at the exploratory data, we looked at and we commented on the uneven distributions of ratings across users and across films. And so there's a risk that models will learn the most from the most frequently
rated films, the most highly active users. I mean, it's the risk, it's the way it works. The less users contribute, the less we know about them, and therefore the less we can predict about their behaviour. But this could mean that less popular films are at a disadvantage. And so
less helpful recommendations are provided to users with less rating history. Movie Lens 1M does contain demographic information, so it talks about the different user groups and we could look at applying that kind of information within our user

groups and examine them.
and use them as part of our recommender system. But the above points do raise questions about fairness and representation and whether the data set reflects the wider population of potential users. So this could be loads of users out there who've never used it yet. And when they come to us as cold users, we have no idea what to recommend to them. So
Is there something that we could do there to make it fairer, to make it more likely to give out the right information?
There are key biases obviously in the movie lens data, so the demographic we've just spoken about, different ages and genders differently represented. Popularity, so some films are rated far more because you get your blockbuster film that everybody goes to see and everybody rates it and you get other films that are just as good but aren't reaching as many people.
And then activity. So again, some users are doing nothing but watch films all day, it would appear, and other users just watch the odd one here and there.
So to take that into account to see how that's working, I looked at the prediction error by user activity. So we have our low, medium and high users. And in the two models, we looked at the error of our predictions. None of them were brilliant.
but we're relatively consistent between the two models here. For error by gender, NCF, again, slightly behind, MS getting slightly better ratings, but again, they're not that dissimilar. And we do see that there is error by gender and it's not as disparate as I would expect it to be between male and female.
And then prediction error by film popularity. So the ones with low popularity and the ones with high popularity, the high popularity have less error and the low popularity have higher error. But again, there's still not very much in it at all.
In terms of deploying this in a real life situation, so things to consider is that machine learning just uses a lot of resources to run generally. And implementing A simpler model is going to be less of a drain on these resources than implementing a more complex model. And as NCF didn't show any real improvement, I'd definitely choose. MF based on my findings so far. So cold start users and films, I've mentioned a couple of times already, but yes, we need to look at what we can do in real world to try and account for those. Either system would require ongoing monitoring, make sure that it was making reasonable
predictions. And we really want to look at those low activity users, the female users and the films with the lower ratings frequency to improve those over time, make

them more correct. Personal data is held against these users, so this is something that you need to consider. It is anonymized,
but it's still probably possible to link this back to specific people if you have the time and the energy to do so. So we need to make sure that the data is properly stored, that GDPR is accounted for, and the similar rules in other countries because this is a worldwide accessible database. And of course, finally, new films are released all the time, so the database needs to be regularly
dated, the model will have to be retrained with the new data and this maintenance would definitely be easier on the traditional model.
My conclusion is that in the case of Movie Lens 1M, which is what I've reviewed here, I've been inclined to use a traditional matrix factorization model for my recommender system for the following reasons. So basically, it has the best results overall. It's easier to build and maintain. It accounts for user and film biases.
which means that the recommendations are a little bit more understandable. And NCF didn't give us improved results, despite all of its extra bells and whistles. The results were extremely close, but in the testing that I did, it did not give us a better result. So there is no argument for deploying that rather than the traditional model.
Here are my references and thank you very much for listening and I hope to do another module again soon.

◉ **Woodcock, Sarah L** stopped transcription